Ceramic Hydrostatic Gel Assembly Improves Cell Viability and Physiological Function of Human Induced Pluripotent Stem Cell-Derived Cardiac Organoids

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Preprint

Outline

- [illegible]
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- [illegible]
- [illegible]
- [illegible]
- [illegible]

Quick Review: What Was Done Last Week?

- **Organization vs Architecture** — implementation vs what the programmer sees
- **Internal organization** of a computer: CPU, memory, I/O and the three buses
  - **Address bus** — [illegible]
  - **Data bus** — bidirectional, [illegible]
  - **Control bus** — [illegible]
- **Memory** [illegible] RAM vs ROM [illegible]
- **Registers** [illegible]
- **Fetch-Decode-Execute** cycle [illegible]
- **First program** [illegible]

Where We Are Heading Today

Lecture 1 and 2	Lecture 3
Basic data type: Variables and Arithmetic Expressions	Basic control flow: Selection and Loop

NASM Listing File (..cont)

How Do We Measure Classifiers?

- The confusion matrix is simply counting how many of the generated cases are classified correctly.
- Measures are calculated from the confusion matrix. For example:

	Predicted Class			
		Yes	No	
Actual Class	Yes	TP	FN	P
	No	FP	TN	N
		P'	N'	All

- Accuracy = (TP + TN)/All
- Error rate = (FP + FN)/All
- Sensitivity = TP/P
- Specificity = TN/N
- Precision = TP/P'
- Recall = TP/P

Key Progress in Data Science & Metrics

- Data science team grew from 3 to 12 members
- Model accuracy improved by 23% year-over-year
- Deployed 4 production ML models serving real-time predictions
- Reduced data pipeline latency from 45 minutes to under 5 minutes

Encoding Text Before Feed: ASCII, UTF

- `str.encode(...)` → `bytes` (text → machine form)
- `bytes.decode(...)` → `str` (back to human form)
- Default encoding: `'utf-8'` (Unicode, variable-length)
- Common alternatives: `'ascii'`, `'latin-1'`, `'cp1252'`
- Error modes: `'strict'` (default), `'ignore'`, `'replace'`, `'backslashreplace'`

Real-Time Offload for Processing Applications

- The offload system from [CHEN11] relies only on CUDA and is unavailable for Windows. Other frameworks, like OpenCL, don't support Windows 8 + ARM. But Windows 8 + ARM is the platform we're targeting.

- Client – the mobile device that is offloading its processing to the server.
- Server – the remote computer that is executing the offloaded processing on behalf of the client.
- **Augmented application** – a mobile application that uses the framework's services.

Why the IBM PC and Intel Architecture?

- We use the **IBM PC based on Intel architecture** because of:
  - [illegible]
  - [illegible]
  - [illegible]
- **iAPX 086** = "Intel Advanced Processor Extensions 86", also called the **8086**
- Used in the **first IBM PC** machines
- **16-bit processor**: registers and connections are 16 bits wide
- [illegible] up to **1 MB** of memory [illegible]

From the 2004 ICT to the 5th

- The **2004 ICT** will take 2004 as the starting point, but actually
- It will be the **5th ICT** in the series
- Each ICT has been organized in a different country
- The **2007 ITU** Plenipotentiary Conference will decide where the 2004 ICT will be held

Exercice 1 (Calcul mental) — Additionner des fractions

Énoncé

Calculer mentalement :

$A = \frac{1}{2} + \frac{1}{4}$; $B = \frac{1}{3} + \frac{1}{6}$; $C = \frac{2}{5} + \frac{1}{10}$; $D = \frac{3}{4} + \frac{1}{8}$; $E = \frac{1}{2} + \frac{1}{3}$.

Solution

$A = \frac{2}{4} + \frac{1}{4} = \frac{3}{4}$; $B = \frac{2}{6} + \frac{1}{6} = \frac{3}{6} = \frac{1}{2}$; $C = \frac{4}{10} + \frac{1}{10} = \frac{5}{10} = \frac{1}{2}$; $D = \frac{6}{8} + \frac{1}{8} = \frac{7}{8}$; $E = \frac{3}{6} + \frac{2}{6} = \frac{5}{6}$.

Méthode

Mettre au même dénominateur, additionner les numérateurs, simplifier si possible.

Register Architecture

General Purpose Registers

[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]

AX = 16 bits

[illegible]	[illegible]

- [illegible] AH, AL, BH, BL, CH, CL, DH, DL
- [illegible]

Index Registers: SI and DI

- SI = Source Index   DI = Destination Index
- They hold memory addresses for data access, not data itself
- Frequently used in string operations (copy, compare, scan a block of bytes)
- They are 16-bit registers and cannot be split into 8-bit halves
- Think of them as pointers to the source and the destination data

Special Registers: IP, SP, BP

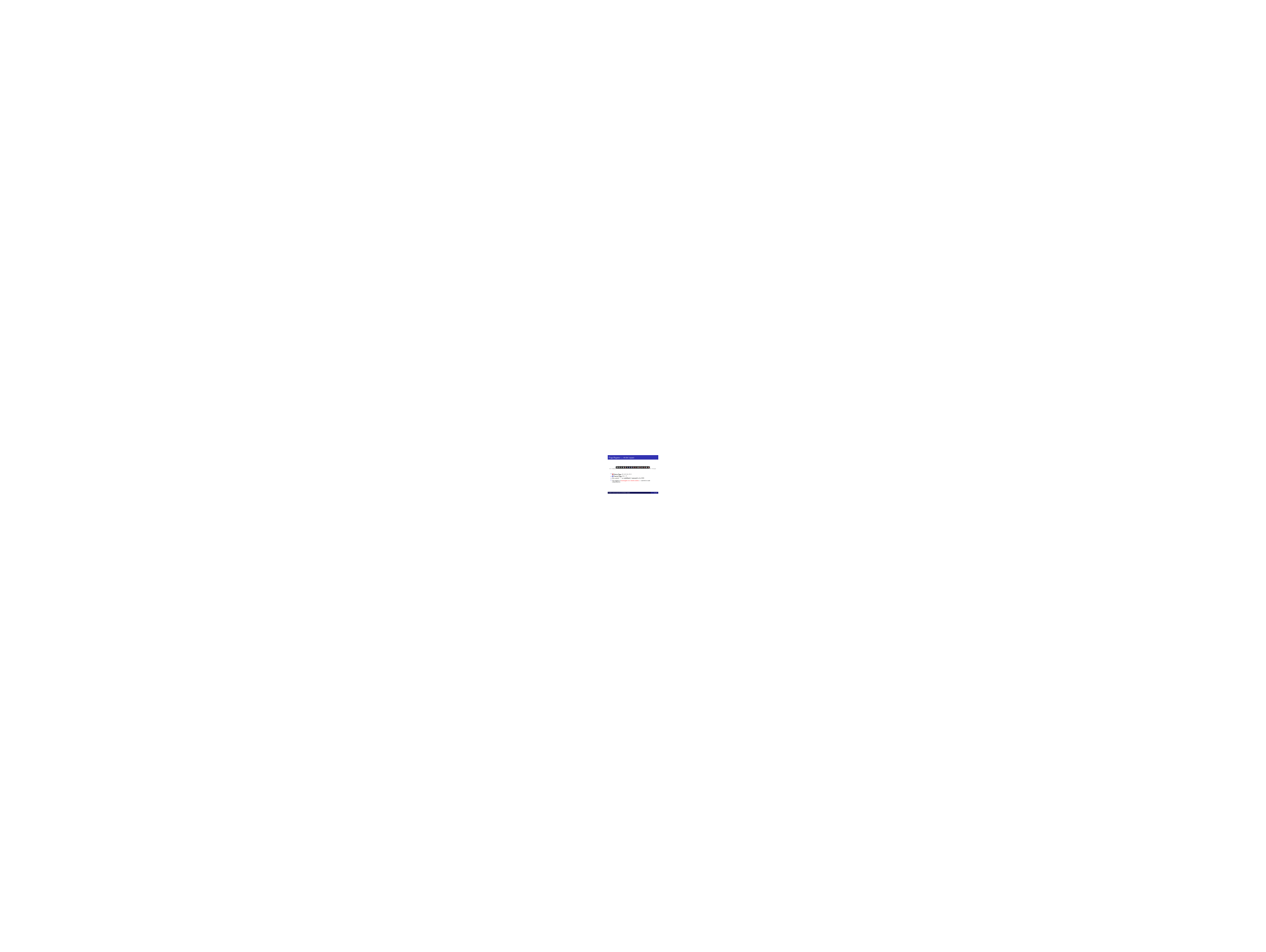
- **IP — Instruction Pointer**
  - Holds the address of the **next instruction**
  - Controlled automatically by the CPU
  - Dangerous to modify directly — it changes the flow of the program
- **SP — Stack Pointer**
  - Points to the **top of the stack**
- **BP — Base Pointer**
  - Used for **stack addressing** (e.g., reaching function parameters and local variables)

The Flags Register

[illegible]	[illegible]	[illegible]

- [illegible]

Page Replacement — the OS chapter



- **Minimum Page** 0 1 2 3 4 5 6
- **Minimum Page** 0 1 2 3
- **Page Fault** 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100
- **Page Hit** 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100

Explore Playlists

1. Each has a **suggestedAt** (meaning decided)
2. Four playlist features:
   - **Top** — Most Played
   - **Hot** — Most Skipped
   - **Cold** — Least Played
   - **Fresh** — Recently Added
3. Every track has a **heading** attribute (**North** of it's previous, **eggplant** of it's previous, etc.)
4. Each playlist has a **suggestedAt** attribute

Augmented Memory Model — Illustration

1. Written sentences (SAM, 2022) could be **fleeting memory signals**: the short-memory system is the subject of the LSTM memory layer (S. Hochreiter & J. Schmidhuber, 1997).
2. Written long-term (LOCOMO) text cannot be **randomly accessible** when the long-term memory is a vector database (P. Lewis et al., 2020).
3. Long-term memory is built and **updated during conversation** (Z. Zhong et al., 2023).

Augmented memory model: Long-term memory is a **knowledge graph** (Y. Hu et al., 2025) that is **updated during conversation** (Z. Zhong et al., 2023). The short-memory system is the subject of the LSTM memory layer (S. Hochreiter & J. Schmidhuber, 1997).

Augmented Balancing Model — Mechanisms

- The central variable is **Perceived Legitimacy** (PL) of the institution. PL is a function of **Procedural Fairness** (PF) and **Outcome Favorability** (OF).
- PF is measured by **Voice** (opportunity to present arguments) and **Neutrality** (consistency of rule application).
- OF is measured by **Proportionality** (sanction severity relative to violation) and **Restorability** (possibility of repairing harm).
- **Cognitive Load** (CL) moderates the PF → PL relationship: high CL reduces the weight of PF in legitimacy judgments.
- **Moral Disengagement** (MD) moderates the OF → PL relationship: high MD reduces the weight of OF in legitimacy judgments.
- **Trust in Institution** (TI) is a lagged predictor of PL.

Hypothesis: Procedural fairness has a stronger effect on perceived legitimacy when cognitive load is high, because individuals rely more on heuristic cues (process) than on effortful evaluation (outcomes).

Exercise Effect — The Two-Way Variables

- Compared with those who were not engaged in exercise, those who did exercise
  - Had lower [BP, SBP, DBP, HR] and BMI as well as lower [TG, HDL, LDL, glucose] levels; CRP
  - Had lower [BP, SBP, DBP, HR] and BMI as well as lower [TG, HDL, LDL, glucose] levels; CRP
- Exercise was associated with a lower risk of [hypertension, diabetes, dyslipidemia, and obesity]
- Exercise was associated with lower [BP, SBP, DBP, HR] and BMI as well as lower [TG, HDL, LDL, glucose] levels; CRP and lower risk of [hypertension, diabetes, dyslipidemia, and obesity]
- Exercise, CRP, and lipid were found positive to be insignificant

Why exercise reduces the risk of hypertension, diabetes, dyslipidemia, and obesity? Exercise is associated with lower [BP, SBP, DBP, HR] and BMI as well as lower [TG, HDL, LDL, glucose] levels; CRP and lower risk of [hypertension, diabetes, dyslipidemia, and obesity]

Physical Address Calculation

Work Organized

1. Survey 2016 studies on what's in practice in **Risk Analysis software**
2. Propose improvements to the RA process and methodology
3. Develop prototype of **new software programmed** for analysis: GUI, input/output, RA Engine, Database
4. Show how improved RMA and software means in everyday results for a RM analyst

Example 2 – Equations – MATH, TEX or LATEX

- Pythagoras's theorem: $a^2 + b^2 = c^2$.
- Euler's identity: $e^{i\theta} = \cos\theta + i\sin\theta$.
- Euler's equation: $V - E + F = 2$.
- Schrödinger's equation: $i\hbar\frac{\partial}{\partial t}\Psi = \hat{H}\Psi$.
- Einstein's field equation: $R_{\mu\nu} - \frac{1}{2}R g_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}$.

Author	Title
Author 1	Paper 1
Author 2	Paper 2
Author 3	Paper 3

Example 1: Hello World Program

1. When the program starts, it will load the main memory (i.e. RAM) with the code for "Hello World!".
2. Within the CPU, a GOTO statement in the program counter will be executed, causing the program to start executing the code at the beginning of the program.

3. An output instruction will then be executed, causing the CPU to send the string "Hello World!" to the output device.
4. The program will then halt, causing the CPU to send the instruction to halt to the CPU.

Original Functions

- The square wave is a 2π-periodic odd function and square(t) = sin(t).
- Fourier series: $\frac{4}{\pi}\sin(t) + \frac{4}{3\pi}\sin(3t) + \frac{4}{5\pi}\sin(5t) + \cdots$
- Will now consider sum $\frac{4}{\pi}\sin(t) + \frac{4}{3\pi}\sin(3t)$ as a **Fourier Transformation**, which is a map $\mathbb{R} \to \mathbb{R}$.
  - $\frac{4}{\pi}\sin(t) + \frac{4}{3\pi}\sin(3t)$: $\mathbb{R} \to \mathbb{R}$
  - $\frac{4}{\pi}\sin(t) + \frac{4}{3\pi}\sin(3t) + \frac{4}{5\pi}\sin(5t)$: $\mathbb{R} \to \mathbb{R}$ (next)
  - $\frac{4}{\pi}\sin(t) + \frac{4}{3\pi}\sin(3t) + \frac{4}{5\pi}\sin(5t) + \frac{4}{7\pi}\sin(7t)$: $\mathbb{R} \to \mathbb{R}$

- Fourier series: $\frac{4}{\pi}\sin(t) + \frac{4}{3\pi}\sin(3t) + \frac{4}{5\pi}\sin(5t) + \cdots$: $\mathbb{R} \to \mathbb{R}$
- Fourier series: $\frac{4}{\pi}\sin(t) + \frac{4}{3\pi}\sin(3t) + \frac{4}{5\pi}\sin(5t) + \cdots$: $\mathbb{R} \to [-1, 1]$

Purchasing Organizers

- Procurement Specialist (PS): Responsible for purchasing activities.
- Purchasing Group (PG): Buyer or group of buyers responsible for purchasing activities.
- Purchasing Organization (PO): Level of the organization that negotiates conditions of purchase with external suppliers.

- Purchasing Organizations can be Centralized, Decentralized, or Combination.

Centralized: Purchasing Organization is responsible for all purchasing activities for all plants. Purchasing Organization negotiates conditions of purchase with external suppliers. Purchasing Organization is responsible for all purchasing activities for all plants.

Logical vs Physical Address

- The **segment:offset** pair is called a **logical address**.
- The 20-bit address is the **physical address** — the real thing on the address bus.
- Logical addressing is only a translation to access physical memory, and you just need: **Many different logical addresses match the same physical address.**

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Thank You

Any Questions?